Short fragment gene sequence splicing APP software

SSGAHG

User's manual

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catalogue

1、 Software description

1.1 Application background

The short fragment gene sequence splicing APP is suitable for researchers engaged in the field of bioinformatics, providing a feasible solution with simple operation and high accuracy of splicing, so as to carry out the analysis and other researches related to genomic information under the conditions of limited time and funds.

1.2 function declaration

(1)SSGAHG.pl

Input: Four files: eg-dp.blast, EG.new.gff, PD.new.gff, PD.new.fa

Output: eg-bp.blast.filter. Sequence information (PD-EG.prechr.gff, PD-EG.prechr.fa) after blast and short fragment gene sequence assembly.

Core function: First, according to the blastp comparison results, the optimal match of each scaffold fragment (eg-bp.blast.filter. blast) is screened, and then the reference chromosome sequence (EG.new.gff) gene is compared, and the pseudo chromosome sequence information is spliced and output.

（2）BlastPlot.pl

Input: Three files: Pa_Eg.blast, Eg.new.gff, PD-EG.prechr.gff

Output: <ϟ1>.png

Core function: Draw PD-EG image

1.3 APP directory display

The Perl source program script is stored in the bin directory

名称	修改日期	类型	大小
.vscode	2025/4/24 14:17	文件夹	
bin	2025/4/24 14:05	文件夹	
README.md	2025/4/24 14:09	Markdown 源文件	0 KB
Uinstall.exe	2025/4/25 20:25	应用程序	90 KB

2、 Hardware and software environment requirements

2.1 Hardware environment requirements

Recommended configuration
CPU: multi-core processor; Memory: more than 512GB;

The development environment can be referenced as follows

系统类型	基于 x64 的电脑
系统 SKU	0B64
处理器	12th Gen Intel(R) Core(TM) i5-12500H，2500 Mhz，12 个内核，

2.2 Software environment requirements

(1) Software environment configuration under Windows system

```
Software tools: Visual Studio code, StrawberryPerl 5.X+
VS code Extensions: Perl, Perl Debugger
```

install cpanm

```
perl -MCPAN -e "install App::cpanminus"
```

Install Perl modules

```
cpanm Perl::LanguageServer
```

VS code Install Perl dependency debugging under the integrated terminal

```
cpanm PadWalker
```

Configure the Perl extension VS code

Ensure that the extension points to the correct Perl interpreter: The Perl interpreter path does not Strawberry-Perl the Perl.exe file path.

(2) Environment configuration under Linux system
install Perl

```
sudo apt install perl
```

install cpanm

```
cpan App::cpanminus
```

3、 Software installation and uninstallation

3.1 install

（1）Windows System
Download and decompress the SSGAHG-V1.0.zip file to the corresponding directory.

（2）Linux system
Unzip the APP folder

```
unzip SSGAHG.zip-d SSGAHG # The compressed package is named SSGAHG.zip
cd SSGAHG
```

Run the SSGAHG.pl script

```
# The script needs to have execution rights
perl SSGAHG.pl
```

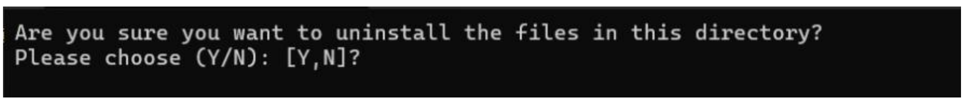
3.2 unload

（1）Windows System

Method 1: Uninstall manually and delete the entire project package.

Method 2: Under the SSGAHG project directory, execute the file Uinstall.exe to uninstall the project files, including the data in the data directory. The following is a picture showing the prompt message of deleting information when executing the Uinstall.exe file to open the PowerShell window. Enter Y and press Enter to execute the deletion operation step by step.

Note: When using method 2, pay attention to back up related data to prevent loss!!!!



```
Are you sure you want to uninstall the files in this directory?
Please choose (Y/N): [Y,N]?Y
Deleted successfully: 新建 DOCX 文档.docx
Deleted successfully: 新建 文本文档.txt
F:\items\SoftwareApply\GeneSeq\Uinstall.bat
另一个程序正在使用此文件，进程无法访问。
Deletion failed: Uinstall.bat
F:\items\SoftwareApply\GeneSeq\Uinstall.exe
拒绝访问。
Deletion failed: Uinstall.exe
Do you want to delete the current folder F:\items\SoftwareApply\GeneSeq\ (Y/N)? Y
F:\items\SoftwareApply\GeneSeq\Uinstall.exe - 拒绝访问。
另一个程序正在使用此文件，进程无法访问。
Folder deletion failed
Total number of deleted files: 2
请按任意键继续. . . |
```

(2) Under Linux system

```
# First, confirm the contents of the directory
ls -l SSGAHG/
# Interactive deletion (file by file confirmation)
sudo rm -ri SSGAHG/
#Or just delete it
sudo rm -rf SSGAHG/
```

4、 instructions to the user

4.1 Windows System

（1）File directory structure

```
SSGAHG/
├── bin/
|    ├── eg-dp.blast
|    ├── PD.new.gff
|    ├── PD.new.fa
|    ├── EG.new.gff
|    ├── ref.fa
|    ├── query.fa
|    ├── BlastPlot.pl
|  └── SSGAHG.pl # Main script file
└── .vscode/
|  └── launch.json # VS Code Debug configuration file
└── README.md          # entry specification
└── Uinstall.exe       # Used to uninstall projects
└── blast-2.16.0+       #blast rely on
```

（2）User operation guide

Install blast dependencies in bin directory, refer to the link: https://ftp.ncbi.nlm.nih.gov/blast/executables/blast+/LATEST/ncbi-blast-2.16.0+-win64.exe

Open cmd in the bin directory and enter the following command:

```
makeblastdb -in ref.fa -dbtype prot -parse_seqids -out egpep
```

```
Building a new DB, current time: 05/01/2025 15:09:08
New DB name:   F:\items\SoftwareApply\GeneSeq-V1.0\GeneSeq\bin\egpep
New DB title:  ref.fa
Sequence type: Protein
Keep MBits: T
Maximum file size: 3000000000B
Adding sequences from FASTA; added 43551 sequences in 2.07449 seconds.
```

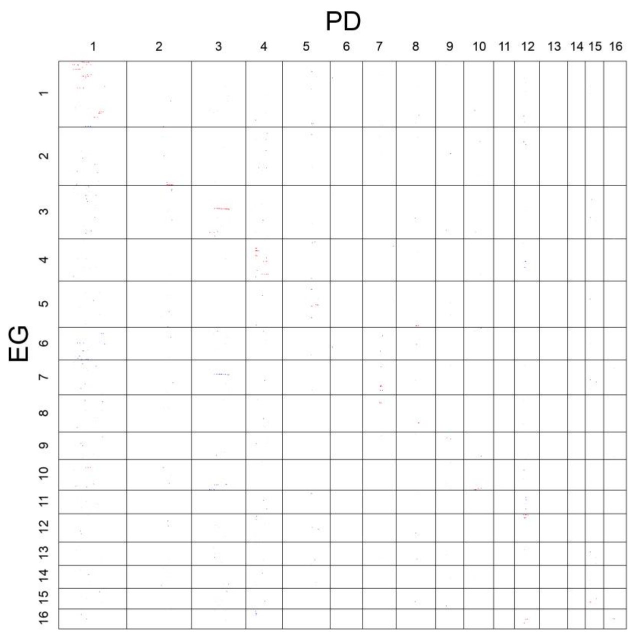
Then enter the following command to generate the Pa_Eg.blast file in the bin directory:

```
blastp -db egpep -query query.fa -task blastp -outfmt 6 -evalue 1e-5 -out Pa_Eg.blast -max_target_seqs 30
```

Enter the perl SSGAHG.pl command in the bin directory, execute the SSGAHG.pl script, and output the PD-EG.prechr.fa and PD-EG.prechr.gff files.

Then enter the following name, execute the BlastPlot.pl script, and generate the PD.EG.blast2dotplot.png file in the bin directory:

```
perl BlastPlot.pl all all PD-EG.prechr.gff EG.new.gff eg-dp.blast order PD EG 1 0 5
```



4.2 Linux system

（1）File directory structure

```
SSGAHG/
├── bin/
│    ├── eg-dp.blast
│    ├── PD.new.gff
│    ├── PD.new.fa
│    ├── EG.new.gff
│    ├── ref.fa
│    ├── query.fa
│    ├── BlastPlot.pl
│  └── SSGAHG.pl  #  Main  script  file
└── Makefile.PL                 Build  the  file
└── README.md                   #  entry  specification
└── blast-2.16.0+               #blast  rely  on
```

（3）User operation guide

Reference link for blast dependency installation under Linux:

https://ftp.ncbi.nlm.nih.gov/blast/executables/blast+/LATEST/ncbi-blast-2.16.0+-x64-linux.tar.gz

First install the pkg-config tool

```
sudo apt-get install pkg-config
```

Install ExtUtils: PkgConfig module

```
sudo cpan install ExtUtils::PkgConfig
```

Install the GD module

```
sudo cpan install GD
```

Replace the files in the bin directory with the corresponding files of the same name and execute the following command.

```
cd SSGAHG # Enter the project directory
bin/SSGAHG.pl #Execute the SSGAHG.pl script in the bin directory
bin/BlastPlot.pl #Execute the BlastPlot.pl script in the bin directory
```

5、 Execution results

5.1 Windows System

Windows In the Visual Studio code development environment of the system, cd + the script path is executed to enter the bin directory of the project, and then perl SSGAHG.pl is executed to execute the script. The execution result is shown in the figure below.

```
DEBUG: Script started
blast filter and rename end!
gff file end
connect scaffold together end!
```

```
3159_368_596_365_1425_281_159_3275_32_580_467_782_287_1912_375_58_3176_1155_1065_329_482_1491_2157_540_55_575_140_172_403_419_2119_358_893_3005_1186_1241_353_75
8_1296_1046_613_1209_1718_2924_1232_2215_676_1668_728_1010_605_626_634_1697_468_963_
1392_1363_1329_1835_220_546_2658_1370_587_601_2479_473_1391_1226_344_875_438_184_3122_2388_1324_44_2929_141_449_2247_1030_274_178_1469_2599_52_2774_2429_1432_39
4_415_608_160_1798_142_1568_203_49_89_538_437_
1913_39_919_979_2589_2904_2831_1988_8_2753_863_1069_7_1217_2996_674_2311_2997_703_1271_1335_1295_671_1511_1075_985_2793_1358_655_969_1289_3270_2788_169_691_2010
_349_2619_3011_376_1227_509_593_1362_846_2443_2502_259_1138_297_2834_313_424_1852_117_829_413_1221_109_323_2646_
scaffold filter end!
fasta file end!
1
2
3
4
5
```

5.2 Linux system

Under Linux system, the perl SSGAHG.pl command is also executed in the perl script directory to execute the script. The execution result is shown in the figure below.

```
root@iZ0jl4i17ae9vu6m6xpnf7Z:/var/GeneSeq-V1.0/GeneSeq/bin# perl GeneSeq.pl
DEBUG: Script started
blast filter and rename end!
gff file end
connect scaffold together end!
```

```
963_
1392_1363_1329_1835_220_546_2658_1370_587_601_2479_473_1391_1226_344_875_438_184
_3122_2388_1324_44_2929_141_449_2247_1030_274_178_1469_2599_52_2774_2429_1432_39
4_415_608_160_1798_142_1568_203_49_89_538_437_
1913_39_919_979_2589_2904_2831_1988_8_2753_863_1069_7_1217_2996_674_2311_2997_70
3_1271_1335_1295_671_1511_1075_985_2793_1358_655_969_1289_3270_2788_169_691_2010
_349_2619_3011_376_1227_509_593_1362_846_2443_2502_259_1138_297_2834_313_424_185
2_117_829_413_1221_109_323_2646_
scaffold filter end!
fasta file end!
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
root@iZ0jl4i17ae9vu6m6xpnf7Z:/var/GeneSeq-V1.0/GeneSeq/bin#
```

```
Usage :  perl this_script querychr sbjctchr querygff1 objctgff2 blastm8 pos_or_order queryinitial sbjctinitial e-value score hitnumber len1 len2 dotpix

some parameters may need to be changed int this_script

outerframe: 2000, 2000
accumulated chro length is 33785
```

After the execution of each script, the file structure under bin directory is shown in the following figure:

Pa_Eg.blast	2025/5/1 15:15	BLAST 文件	495 KB
egpep.pdb	2025/5/1 15:09	Program Debug ...	2,572 KB
egpep.phr	2025/5/1 15:09	PHR 文件	5,272 KB
egpep.pin	2025/5/1 15:09	PIN 文件	341 KB
egpep.pjs	2025/5/1 15:09	PJS 文件	1 KB
egpep.pog	2025/5/1 15:09	POG 文件	171 KB
egpep.pos	2025/5/1 15:09	ProcessOn Docu...	979 KB
egpep.pot	2025/5/1 15:09	Microsoft Power...	511 KB
egpep.psq	2025/5/1 15:09	PSQ 文件	20,300 KB
egpep.ptf	2025/5/1 15:09	PTF 文件	489 KB
egpep.pto	2025/5/1 15:09	PTO 文件	171 KB
PD-EG.prechr.fa	2025/5/1 15:01	FA 文件	1 KB
PD-EG.prechr.gff	2025/5/1 15:01	GFF 文件	1,281 KB
eg-dp.blast.filter.blast	2025/5/1 15:00	BLAST 文件	1,207 KB
SSGAHG.pl	2025/5/1 14:59	Perl 源文件	6 KB
ref.fa	2025/4/30 23:01	FA 文件	23,889 KB
query.fa	2025/4/30 23:01	FA 文件	21,736 KB
Arial.ttf	2025/4/27 10:00	TrueType 字体文件	756 KB
BlastPlot.pl	2025/4/27 10:00	Perl 源文件	14 KB
PD.new.fa	2025/4/24 20:41	FA 文件	426,266 KB
eg-dp.blast	2025/4/24 20:38	BLAST 文件	41,128 KB
EG.new.gff	2025/4/24 17:04	GFF 文件	1,180 KB
PD.new.gff	2025/4/24 17:04	GFF 文件	1,310 KB
blast-2.16.0+	2025/4/30 21:27	文件夹	